Table 4. Authentication criteria for salivary gland cancer patient-derived organoids

Criterion	Recommended methods	Assessment and reporting
Histomorphology	Hematoxylin and eosin (H&E) staining; side-by-side comparison with the parental tumor.	Compare characteristic architecture, cytology and subtype-defining patterns. Document discrepancies and sampling limitations.
Subtype-defining driver alterations	Reverse-transcription PCR, fluorescence in situ hybridization (FISH), or fusion-capable sequencing, as appropriate.	Confirm retention of alterations identified in the parental tumor (e.g., MYB–NFIB / MYBL1–NFIB in ACC, CRTC1/3–MAML2 in MEC, or ETV6–NTRK3 in secretory carcinoma).
Mutation and copy-number profile	Targeted DNA sequencing, whole-exome sequencing, and/or copy-number analysis according to tumor subtype and available tissue.	Assess concordance with informative parental-tumor alterations, including amplifications and copy-number changes; report newly detected or lost alterations.
Transcriptomic fidelity	RNA sequencing and/or focused expression profiling when relevant to the study purpose.	Assess preservation of lineage- and tumor-associated transcriptional programs; report differences with the parental tumor, where matched data exist.
Passage history and stability	Record passage number; repeat selected morphology and molecular tests after prolonged expansion or before functional screening.	Specify passage numbers at authentication and drug testing; document culture-associated drift or clonal selection and reassessment intervals.
Tumor identity and culture purity	Subtype-appropriate immunohistochemistry / immunofluorescence, morphology and molecular confirmation.	Distinguish neoplastic epithelium from residual normal ductal, basal/myoepithelial and stromal populations; document contamination or overgrowth.

Assay selection should be tailored to tumor subtype, intended use and the defining alterations present in the matched parental specimen. No single sequencing modality is mandatory for every culture.

Abbreviations: ACC, adenoid cystic carcinoma; MEC, mucoepidermoid carcinoma; PCR, polymerase chain reaction; FISH, fluorescence in situ hybridization.